A Data-led Approach To Customer Retention

Telecom Customer Churn Prediction

Business question

Can we identify customers at risk of leaving early enough to focus limited retention resources where they may have the greatest impact?

Project focus: predict churn, understand model trade-offs, and support smarter retention decisions.

BUSINESS CASE • MACHINE LEARNING • CUSTOMER VALUE



PREDICT → PRIORITISE → RETAIN

The business problem: limited budget, costly churn

The challenge

Customers can leave for competitors. The company cannot give every customer the same level of retention attention, so it needs a practical way to prioritise who may be at risk.

Revenue at risk

A churned customer can mean lost recurring revenue and future customer value.

Limited capacity

Discounts, service calls and offers cost money and staff time.

What the model can enable

- **Prioritize:** rank customers by estimated churn risk.
- **Focus:** direct limited outreach to a manageable group.
- **Learn:** measure which interventions reduce churn.

Business value to test — **Fewer avoidable losses • better-targeted retention spend • clearer visibility of at-risk customers**

Cost note: financial impact has not yet been quantified; estimate it using customer lifetime value, intervention cost and measured retention uplift.

The data: customer and service signals

7,043
Customer records
From Kaggle

21
Columns
Customer, service, billing etc

1
Target
Churn: 1 = yes, 0 = no

Key features tested

- **ALL —** Categorical Columns were converted into numerical and was encoded. The numerical Columns was standardized

Data-quality checks

- **Missing values:** no null values. Identified 11 blank strings in “Total Charges Feature.
- **Data types:** Total Charge initially had a string data type but was changed to numeric
- **Duplicates:** No Duplicates recorded
- **Categories:** All label was consistent
- **Outliers:** 0 Potential Outliers for Numerical and Categorical Features

Source: Telco Customer Churn CSV used in the project. Current analysis focuses first on five selected categorical predictors.

Approach and results: Random Forest leads on key retention metrics

Approach

- **Problem type:** binary classification — predict churn / no churn.
- **Models compared:** Logistic Regression, Decision Tree ,Random Forest and KNN
- **Evaluation:** same train/test split; accuracy, precision, recall, F1, ROC-AUC and
- **Leakage control:** split first; fit preprocessing on training data only.

Test scores

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Decision Tree	71.7%	46.8%	48.1%	47.4%	64.1%
K-Nearest Neighbours	76.4%	55.3%	57.5%	56.4%	79%
Random Forest	78.1%	61.10%	48.1%	53.8%	81.80%
Logistic Regression	**80.6%**	**65.7%**	**55.9%**	**60.4**	**84.2**

Current leading Algorithm: Logistic Regression

Recommendation: pilot the model, measure business impact

Recommendation

Prioritise at-risk customers: Use the Logistic Regression model to identify customers who are likely to churn and target them with retention offers, personalised support, or contract incentives.

Focus on customer retention: Investigate customers with short-term contracts, limited-service support, or other characteristics associated with churn, and develop targeted strategies to improve their satisfaction and loyalty.

Monitor model performance: Regularly evaluate the model using precision, recall, F1-score, and ROC-AUC. Although Logistic Regression achieved 80.6% accuracy and an ROC-AUC of 84.2%, it still failed to identify some customers who churned.

Test and improve retention campaigns: Run a pilot campaign targeting customers flagged as high risk. Track retention rates, campaign costs, and revenue saved to determine whether the model helps reduce churn and improves the return on investment.

Next steps

1. Fine-tune the Logistic Regression model to improve churn prediction.
2. Adjust the prediction threshold to identify more at-risk customers.
3. Run a pilot retention campaign targeting customers predicted to churn.
4. Monitor churn rates, retention costs, and revenue saved to measure business impact.

Risk: The model may incorrectly classify customers, leading to missed churners or wasted retention spending; its predictions should be validated through real-world testing.